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Problem Solving Protocol

TP-LMMSG: a peptide prediction graph neural network incorporating flexible amino acid property representation

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Abstract

Bioactive peptide therapeutics has been a long-standing research topic. Notably, the antimicrobial peptides (AMPs) have been extensively studied for its therapeutic potential. Meanwhile, the demand for annotating other therapeutic peptides, such as antiviral peptides (AVPs) and anticancer peptides (ACPs), also witnessed an increase in recent years. However, we conceive that the structure of peptide chains and the intrinsic information between the amino acids is not fully investigated among the existing protocols. Therefore, we develop a new graph deep learning model, namely TP-LMMSG, which offers lightweight and easy-to-deploy advantages while improving the annotation performance in a generalizable manner. The results indicate that our model can accurately predict the properties of different peptides. The model surpasses the other state-of-the-art models on AMP, AVP and ACP prediction across multiple experimental validated datasets. Moreover, TP-LMMSG also addresses the challenges of time-consuming pre-processing in graph neural network frameworks. With its flexibility in integrating heterogeneous peptide features, our model can provide substantial impacts on the screening and discovery of therapeutic peptides. The source code is available at https://github.com/NanjunChen37/TP_LMMSG.

Keywords: peptide prediction; antimicrobial peptides; graph neural network; transfer learning.

Introduction

The discovery of peptide therapeutics has always been a focal point of drug development. Distinguished from small molecule drugs and biological agents, peptide drugs capture researchers' attention by their unique advantages [1, 2]. In recent years, the progress in synthesis, modification, and analytical technologies have propelled therapeutic peptide research into a new era [2]. Concurrently, the continuous evolution of machine learning models has played a pivotal role in accelerating the peptide drug discovery pipeline. Alongside the widely acknowledged antimicrobial peptides (AMPs) prediction models, the field has witnessed the successive emergence of antiviral peptides (AVPs) and anticancer peptides (ACPs) prediction models [3–5]. The AMPs have attracted widespread attention due to their broad-spectrum antibacterial activity [6, 7]. Specifically, AMPs belongs to a class of short peptides (around 6 to 100 amino acids) involved in the innate immune response among lives [8, 9]. The general therapeutic mechanism of AMPs is to bind the surface of microorganisms through the attraction effect between positive and negative charges. Specifically, most AMPs have net-positive charge, while bacterial membrane has negative charge. The interaction can directly or indirectly destroy the integrity of microbial membrane, thereby affecting the life cycle and proliferation of the microorganism [10, 11]. Moreover, evidence also suggests that the 3D structure of AMPs may also affect the antimicrobial activity of peptides [12, 13]. In addition to natural AMPs, it has been proven promising to discover new synthetic AMPs using generative models and synthetic workflows [14, 15]. Different from small molecules,

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peptide therapies mainly take advantage of higher specificity and lower toxicity. Hence, peptide drugs with specific targets are also adapted as antifungal, antiviral, antiparasitic or even anticancer clinical strategies [8, 16, 17].

To meet the increasing demand for model training and high-throughput screening of bioactive peptides, various public databases have been established. These databases provide different types of peptide information, allowing researchers to design machine learning models based on different objectives. The APD3 [9], dbAMP [18] and DRAMP [19] databases collect available AMP data and classify those AMPs into different categories based on their targets. Moreover, the experimental data and derived data, such as inhibition activity and molecular properties, are also included in these databases. Meanwhile, the databases such as CAMP [20], ADAM [21] and SATPdb [22] provide abundant peptide structural information. In 2021, several experimentally validated datasets were investigated [3], including APD3 [9], CAMP [20], dbAMP [18], DRAMP [19], LAMP [23] and YADAMP [24]. Based on the AMPs records in these datasets, a fusion dataset named XUAMP is constructed to thoroughly evaluate the performance of AMP identification models [3]. Meanwhile, the CD-HIT tool is utilized for rigorous sequence filtering and sampling to construct the independent test dataset [3]. The following model evaluations demonstrate that there is a potential for further enhancement in current AMP prediction models. In addition to AMPs, the APD3, CAMP, DRAMP and LAMP databases also contain diverse peptide information, including AVPs, ACPs and other theraputic peptides that target at different diseases.

Benefit from the peptide information in the aforementioned datasets, multiple feature representation approaches have also been proposed. Besides the commonly used one-hot encoding and multiple sequence alignment (MSA) encoding, incorporating amino acid physicochemical properties can also effectively improve model performance [25, 26]. Meanwhile, the use of positional encoding, amino acid composition and pseudo amino acid composition features was shown to enhance the model performance by incorporating detailed composition information of amino acids [25, 27, 28]. As for the prediction models, the random forest model amPEPpy demonstrates good predictive performance in distinguishing AMPs [29]. Recently, there has been an increasing trend in utilizing deep learning frameworks to distinguish AMPs and non-AMPs [30, 31]. Besides the composition features, the recently published AMPpred-MFA model incorporates dipeptide and k-mer composition as additional input features. Although these methods combine the position, composition and physicochemical properties of peptides, they do not leverage the structural information of the peptide. To tackle this, the sAMPpred-GAT model has been proposed. The model utilizes the protein structure predicted by trRosetta for AMP identification [32, 33]. Except for AMP prediction, some works have transferred the successful experience of predicting AMPs to other types of peptide identification tasks, and have developed prediction models specifically for AVP and ACP [4, 5]. Although sAMPpred-GAT is a promising and innovative attempt, there are still limitations in its practical application. First, the model only utilizes one-hot encoding, positional encoding, hidden Markov models (HMMs) encoding and position-specific scoring matrix (PSSM) encoding as the node features. Two of these features (HMM and PSSM) solely rely on MSA and protein sequence databases [34, 35]. This indicates that the model lacks explicit representations of amino acid physicochemical properties. The two rounds of MSA also implies that the bottleneck in the model's runtime lies in the feature extraction stage (e.g. around 15 s on single sample). Additionally, its complete deployment requires the download of the entire nonredundant protein sequence database (nrdb, 0.5 TB). Therefore, it is crucial to improve the effectiveness and storage occupancy of the model.

In this study, we aim to develop a peptide prediction model, which not only predicts AMPs, but also generalizes to predict other types of peptides (i.e. AVPs and ACPs). Moreover, we aim to reduce the storage occupancy and feature encoding time in graph neural network applications. So far, protein embeddings have been demonstrated to be successful at both sequence and amino acid levels [36, 37]. Meanwhile, the hierarchical multi-scale residual network (HMSRN) has been proved promising in protein–RNA interaction prediction tasks [38]. Therefore, we adapt multiple pre-trained language models to obtain the amino acid embeddings. The HMSRN is then applied to extract informative features among the language model embeddings. Subsequently, we concatenate the extracted features into each node of peptide graph for graph learning. We name our approach as the Therapeutic Peptide Language Model Multi-Scale Graph neural network (TP-LMMSG). The experimental results reveal that our model outperforms the state-of-the-art (SOTA) models in multiple evaluation metrics. Remarkably, these achievements are attained without employing HMM features and rely solely on the lightweight nonredundant protein sequence database 90 (nrdb90) [39]. In addition to AMP prediction, we further generalize our model to the prediction tasks of AVPs and ACPs. Our model also exhibits remarkable performance, surpassing existing models in identifying the other types of peptides with imbalanced datasets.

Material and methods

Datasets

To ensure a fair evaluation and enable an objective comparison with previously proposed models, we primarily utilize the seven AMP datasets provided in previous literatures (i.e. XUAMP, APD3, CAMP, dbAMP, DRAMP, LAMP and YADAMP [3, 9, 18–20, 23, 24, 32]). Specifically, the XUAMP dataset is integrated from multiple datasets including APD3, DRAMP, LAMP, CAMP, dbAMP and YADAMP, resulting in 5536 AMPs/nonAMPs in the training set, and 1536 AMPs/nonAMPs in the independent test set. We choose to benchmark on this dataset as the primary evaluation. Besides the XUAMP dataset, we also conduct evaluations on the other six datasets mentioned above (Table 1, Figure S1). Specifically, each independent test dataset of the six datasets has undergone strict filtering rules, where CD-HIT tool is applied to ensure no sequence shares more than 40% identity in the test set [3, 40, 41]. In this manner, we can rigorously evaluate the performance and make a thorough comparison with previously proposed methods. As for the AVPs and ACPs datasets, the number of positive samples is considerably fewer than that for AMPs [17]. To ensure a fair comparison, we adapt the dataset utilized in previous publications, which are FFMAP AVPs dataset and ACP-ML ACPs dataset, respectively (Table 1) [4, 5].

Methodology

The overview of TP-LMMSG model

The architecture of TP-LMMSG consists of five parts (Figure 1). Firstly, the predictive amino acids connectivity is obtained with trRosetta to construct basic peptide graph (Figure 1A). Secondly, the basic amino acid level features are extracted by conducting one-hot encoding, position encoding, PSI-BLAST and HHBlits on

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Table 1. Peptide sequence statistics of the nine datasets involved in this study

Dataset name	Train dataset		Test dataset	
	Positive	Negative	Positive	Negative
APD3	5251	5188	494	494
CAMP	5504	5394	203	203
dbAMP	5197	5166	522	522
DRAMP	4961	4550	1408	1408
LAMP	4814	4824	1054	1054
XUAMP	5536	5536	1536	1536
YADAMP	5431	5306	324	324
AVP	2762	10 089	691	2522
ACP	487	1479	135	360

peptide sequences. We consider the lightweight graph construction method, where running HHBlits is optional (the portions with dash lines in Figure 1A). These features are assigned as the initial graph node features (Figure 1B). For the third step, the SeqVec and ProtTrans pre-trained protein language models are used to obtain the amino acids level embeddings, including the hidden representation of hydrophilia, electrical properties and polarity [36, 37]. Specifically, the pretrained SeqVec model and PortTrans model are adapted to extract the biophysical embeddings of each amino acid in the peptide sequence [36, 37]. For the output of each pre-trained model, an HMSRN is utilized to extract the physicochemical information at different scales (Figure 1C). In the last two steps, the latent representation is integrated for each amino acid in the peptide graph (Figure 1D). A graph attention network (GAT) followed by fully connected layers is proposed to predict whether the input peptides are positive samples or negative samples (Figure 1E).

Connectivity prediction and basic feature encoding

To obtain the structure information of peptides, the trRosetta model is utilized to obtain the distances between the βC of amino acids based on peptide sequences (Figure 1A) [33]. Once the distance is obtained, the pairwise connectivity of amino acids can be further determined. By aggregating all connectivity information, the adjacent matrix A is defined as follows:

$$A_{u,t} = \begin{cases} 1, & \text{if } D_{u,t} \le D_{\text{th}} \\ 0, & \text{otherwise} \end{cases}, \tag{1}$$

where $D_{u,t}$ denotes the distance between amino acid u and t, and D_{th} is the distance threshold. Since the choice of the distance thresholds D_{th} have an impact on the predictive accuracy of the model [42], we set different thresholds (i.e. D_{th}=[10Å, 15Å, 20Å]) to be evaluated for accurate graph representation.

As the *de facto* standard, the three basic feature extraction methods on amino acids, including one-hot encoding, positional encoding and PSSM encoding, are applied. As for one-hot encoding, given a 20-dimensional vector, the value of 1 is assigned at different dimensions to represent the each amino acid. This yield an encoding matrix with a dimension of $L \times 20$. Moreover, as for positional encoding, Sinusoidal positional encoding sourced from Transformer model is adapted [43]:

$$\begin{cases} PE_{(pos,2i)} = \sin\left(pos/b^{2i/d_{\text{model}}}\right), \\ PE_{(pos,2i+1)} = \cos\left(pos/b^{2i/d_{\text{model}}}\right), 0 \le i \le 10 \end{cases}. \tag{2}$$

In this case, constants b and d are set to 1000 and 20 to maintain consistency [32]. For PSSM encoding, the PSI-BLAST tool is adapted to obtain the MSA results for each sequence based on the nrdb90 [35]. The PSSM features are then derived, where the frequency of each amino acid at each sequence position is recorded. For the sequences where MSA results cannot be generated, we employ the BLOSUM62 matrix for padding in the features matrix. The positional residue match score can be extracted according to the equation

$$score_{ij} = 2^{-0.001\times v_{ij}} (i \in [1, L],\ j \in [1, 20]), \tag{3}$$

where v refers to the amino acid frequency, and i and j indicate the position and name of amino acid on the peptide sequence with a total length of L, respectively. As for the optional HMM feature, HHBlits tool is used to conduct MSA, based on databases such as Uniclust30 [34]. The MSA data can also be conserved to HMM features based on equation 3. When HHBlits is used, its MSA results could be treated as the input data for peptide structure prediction [33]. By the end of this step, the amino acids connectivity and the basic features will facilitate the initialization of peptide graph (Figure 1B).

Protein language model based embeddings extraction

In our lightweight work frame where HMM feature is absent, the sparse node feature representation becomes even scarcer. Furthermore, the shrinkage on nrdb for PSSM encoding poses challenges in accurately representing amino acid features. Therefore, we utilized multiple pre-trained protein language models to obtain informative amino acid feature embeddings (Figure 1C) [36, 37]. First, the pretrained SeqVec model ('sv' in the following description) is adapted [36]. By training EMLo on the UniRef50 dataset, SeqVec can generate continuous embeddings at amino acid level. In this step, the peptide sequences are fed into the pre-trained SeqVec model to obtain amino acid-level embeddings for each sequence. As a result, the model outputs a vector with 1024 dimensions for each amino acid. Due to the varying lengths of sequences, we extract embeddings according to the maximum sequence length (L_{max}). For sequences shorter than L_{max}, padding operation was applied at the end of the sequence. As a result, the embeddings matrix for each sequence is $\mathbf{X}_{lm} \in \mathbb{R}^{L_{max}\times 1024}$, where $lm = \text{sv}$.

We also utilize the PortTrans model, a series of protein language models pre-trained based on BERT, Albert, Electra and T5 [37]. We select three lightweight pre-trained models as the methods for amino acid-level feature extraction, which are 1. ProtT5-XL-U50, 2. ProtT5-XL-BFD and 3. ProtBert-BFD. The models will be labeled as 'uni', 'bfd' and 'brt' in the following description. Similar to the SeqVec model, the output embeddings for each sequence is $\mathbf{X}_{lm} \in \mathbb{R}^{L_{max}\times 1024}$, where $lm \in \{\text{uni}, \text{bfd}, \text{brt}\}$. To enable the downstream model to accept the output embeddings from a flexible number of language models, the output embeddings of each language model is stored separately.

Flexible HMSRN

After obtaining the amino acid level embeddings, we adapt the HMSRN architecture to extract features at the amino acid level (Figure 1C). The aim of utilizing the HMSRN is to capture node features comprehensively while alleviating the gradient vanishing problem [44]. In this work, the HMSRN is optimized to receive a varying number of language model embeddings and conduct feature concatenation at the amino acid level.

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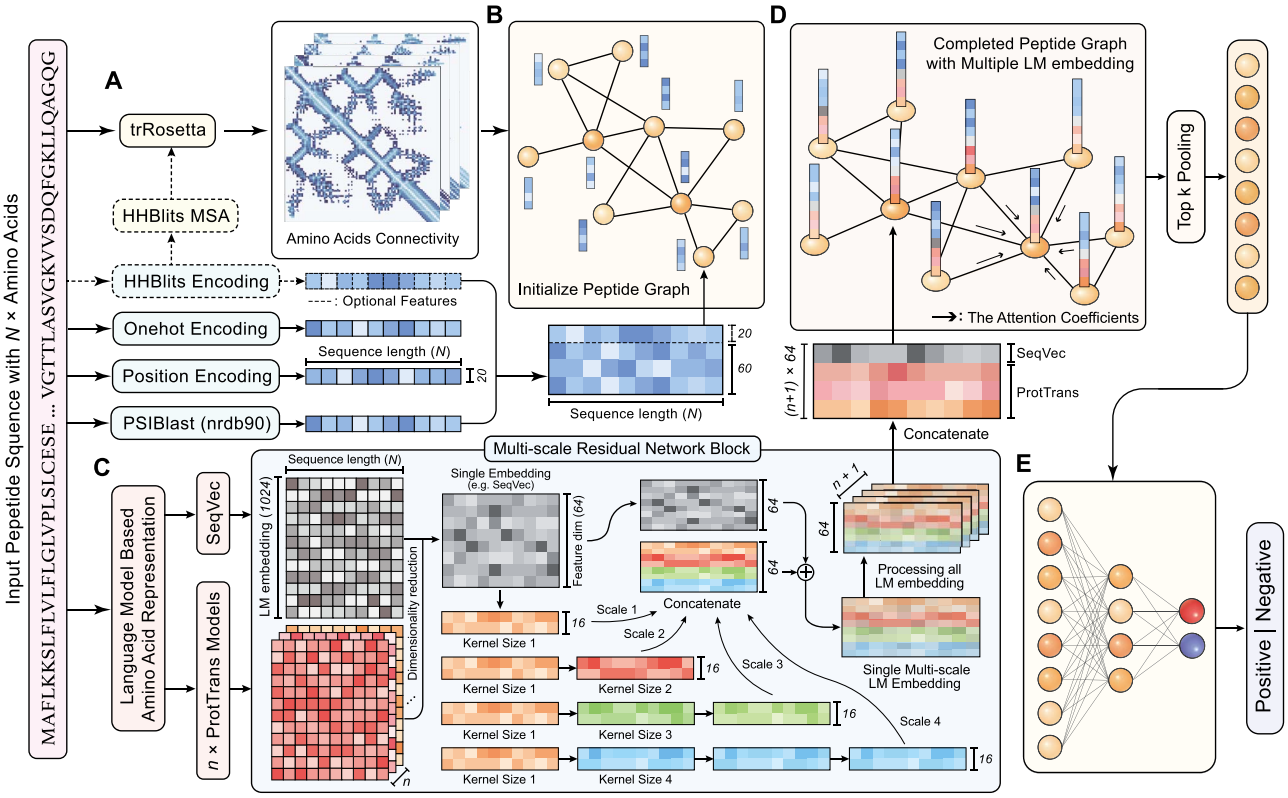

Figure 1. Overview of the proposed workflow. The peptide prediction model TP-LMMSG is divided into five parts: (**A**) Applying basic encoding methods and use trRosetta for amino acid connectivity prediction. (**B**) Peptide graph construction according to βC connectivity and basic amino acid encoding. (**C**) Using flexible number of pre-trained language models to obtain amino acid node level embeddings. Then extract the informative feature through multiple MRSN modules. (**D**) Aggregating amino acid features extracted by language models into peptide graphs. (**E**) Applying the GAT and fully connected layers for peptide property prediction.

The HMSRN architecture consists of three parts. Firstly, a two-layer CNN module was employed to perform preliminary feature extraction on each amino acid while ensuring consistent feature dimensions. The feature extraction is conducted based on each language model embeddings. Mathematically, $\mathbf{X}_{lm} = [\mathbf{x_1}, \mathbf{x_2}, \mathbf{x_3}, \cdots, \mathbf{x_l}]$, $\mathbf{X}_{lm} \in \mathbb{R}^{l \times 1024}$ represents the single language model embeddings for each peptide sample with maximum sequence length l. A 1D-convolution operation is applied to the features of each amino acid vector $\mathbf{x_i}$, as follows:

$$F_C(\mathbf{x}_i) = \mathrm{ReLU}(\mathrm{BN}(\mathrm{Conv1D}(\mathbf{x}_i))), i \in \{1, 2, \ldots, l\} \tag{4}$$

$$\tilde{\mathbf{X}}_{lm} = F_C(\mathbf{X}_{lm}). \tag{5}$$

Afterwards, our model adapts k HMSRN core structures, where k is the number of language model embeddings ($n+1$ in Figure 1). Each HMSRN core structure $H(\cdot)$ contains four CNN modules $F_{C_i^j}(\cdot)$ with different scales for amino acid feature extraction. The $H(\cdot)$ corresponds to certain language model embeddings receives $\tilde{\mathbf{X}}_{lm}$ as input. For each scale, sequential CNN modules with specific kernel size is used to capture the features on each amino acid, as follows:

$$F_{C_i^j}(\tilde{\mathbf{X}}_{lm}) = \mathrm{ReLU}\left(\mathrm{BN}\left(\mathrm{Conv1D}(\tilde{\mathbf{X}}_{lm})\right)\right), i \in \{1, 2, 3, 4\}, j \leq i. \tag{6}$$

Here $F_{C_i^j}(\cdot)$ denotes the j-th sequential module in the i-th scale. Therefore, the combination of sequential CNN modules for certain scale can be denoted by combining all sub-modules:

$$H_i(\tilde{\mathbf{X}}_{lm}) = \prod_i \left(F_{C_i^j}(\tilde{\mathbf{X}}_{lm})\right), i \in \{1, 2, 3, 4\}, j \leq i. \tag{7}$$

To concatenate the output from each scale, the output for the $H(\cdot)$ can be denoted as

$$\hat{\mathbf{X}}_{lm} = \left| H_1(\tilde{\mathbf{X}}_{lm}), H_2(\tilde{\mathbf{X}}_{lm}), H_3(\tilde{\mathbf{X}}_{lm}), H_4(\tilde{\mathbf{X}}_{lm}) \right|. \tag{8}$$

A residual mechanism is further used to fuze the output and the input feature of $H(\cdot)$:

$$\mathrm{Residual}(\tilde{\mathbf{X}}_{lm}) = \tilde{\mathbf{X}}_{lm} + \hat{\mathbf{X}}_{lm}. \tag{9}$$

By processing all k language model embeddings, the feature for each peptide sequence can be denoted as

$$\tilde{\mathbf{X}}'_{lm} = |\mathrm{Residual}_1(\tilde{\mathbf{X}}_{lm}), \mathrm{Residual}_2(\tilde{\mathbf{X}}_{lm}), \ldots, \mathrm{Residual}_k(\tilde{\mathbf{X}}_{lm})|. \tag{10}$$

From a broad perspective, the features generated by k individual language models undergo their dedicated HMSRN processing. Then the output features from each HMSRN blocks are concatenated in corresponding to each amino acid of peptide, facilitating the precise representation for each sequence.

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The architecture of graph attention neural network

In this step, the extracted amino acid level features are integrated into each node of the molecular graph. As for the original peptide graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V} = \{\mathbf{v}_i \mid i \in [1, n]\}$ represents the set of nodes. The $\mathbf{v}_i \in \mathbb{R}^d$ denotes the concatenated basic amino acid level feature aforementioned with a dimension d. After integrating the language model features, the node representation is denoted as $\mathcal{V} = \{\tilde{\mathbf{v}}_i \mid (i \in [1, n])\}$, where $\tilde{\mathbf{v}}_i = \left|\mathbf{v}_i, \tilde{\mathbf{x}}'_i\right|$, and $\tilde{\mathbf{v}}_i \in \mathbb{R}^{\tilde{d}}$. Here $\tilde{d} = d + 64k$, denoting the updated node feature dimensions. $\mathcal{E}$ denotes the edge attribute between a pair of connected amino acids in the peptide graph.

The feature for the t-th graph attention layer is a set of node features $\mathcal{H}^{(t)} = \left\{\mathbf{h}_1^{(t)}, \mathbf{h}_2^{(t)}, \ldots, \mathbf{h}_n^{(t)}\right\}, \mathbf{h}_i^{(t)} \in \mathbb{R}^{d_t}$. n denotes the sequence length of the certain peptide graph, and d_t denotes the dimension of features in the graph attention layer. Based on the node feature of neighbor nodes i and j, an attention coefficient can be calculated to measure the significance of node j to node i, as follows:

$$\alpha_{i,j} = \frac{\exp\left(\text{LeakyReLU}\left(\mathbf{a}_s^\top \mathbf{W}_s \mathbf{x}_i + \mathbf{a}_t^\top \mathbf{W}_t \mathbf{x}_j\right)\right)}{\sum_{k \in \mathcal{N}(i) \cup \{i\}} \exp\left(\text{LeakyReLU}\left(\mathbf{a}_s^\top \mathbf{W}_s \mathbf{x}_i + \mathbf{a}_t^\top \mathbf{W}_t \mathbf{x}_k\right)\right)}, \tag{11}$$

where $\mathbf{a} \in \mathbb{R}^{2d_t}$ denotes a weight vector, and $\mathbf{W}_s$ and $\mathbf{W}_t$ denote the learnable weight matrices, respectively. After all, the output feature for each node can be computed by aggregating over all neighbor nodes:

$$\mathbf{h}_i^{(t+1)} = \alpha_{ii} \mathbf{W}_s \mathbf{h}_i + \sum_{j \in \mathcal{N}(i)} \alpha_{ij} \mathbf{W}_t \mathbf{h}_j. \tag{12}$$

We denote the output of the GAT layers as $\mathcal{H}$. Following the top-k pooling operation, the unified peptide feature matrix is passed through a series of fully connected layers:

$$\mathbf{z} = \mathbf{W}_f \mathcal{H} + \mathbf{b}. \tag{13}$$

Here $\mathbf{W}_f$ and $\mathbf{b}$ denote the learnable weight and bias of the fully connected layer, respectively. For the purpose of discriminating between AMP and non-AMP, we apply **softmax** to the output of the final fully connected layer, yielding the final predictions:

$$\mathbf{y} = \sigma(\mathbf{z}). \tag{14}$$

We labeled the positive samples and negative samples as 1 and 0, respectively. Therefore, y_i denotes the prediction probability of the two classes. The details regarding the implementation and training of the model is attached to the Supplementary Materials.

Evaluation metrics

The evaluation metrics used in this work are listed in equation 15. These metrics allow us to objectively compare the performance of our model with the SOTA models. For the imbalanced datasets ACP and AVP, we also compute the Area Under the Precision-Recall (AUPR) curve.

$$\begin{cases} ACC = \dfrac{TP + TN}{TP + TN + FN + FP} \\ F1\ Score = \dfrac{TP}{TP + \frac{1}{2}(FP + FN)} \\ MCC = \dfrac{TP \times TN - FP \times FN}{\sqrt{(TP + FN)(TP + FP)(TN + FP)(TN + FN)}} \\ Sn = \dfrac{TP}{TP + FN} \\ Sp = \dfrac{TN}{TN + FP} \\ AUC: \text{ Area under the ROC Curve} \\ AUPRC: \text{ Area Under the Precision-Recall Curve} \end{cases} \tag{15}$$

Results

Performance comparison with other methods

We first compare the predictive performance of both the lightweight and heavyweight model with the other 13 methods. The frequently adopted models for performance comparison include AMPscannerV2, iAMP-2L, ADAM-SVM/HMM, ampir, MLAMP, AMPlify, AMPEP, AMPfun and AMPpred-EL, which are comprehensively evaluated in the previous studies [3, 45]. We further replicate other SOTA AMP prediction models, including amPEPpy, AMPpred-MFA and sAMPpred-GAT. We follow the data processing protocols in the previous benchmark experiments to avoid bias [3, 32].

The experiment results of six evaluation metrics on the XUAMP are listed in Table 2 and Figure S2. For both lightweight and heavyweight TP-LMMSG models, only nrdb90 is used to obtain the PSSM feature. The results turn out that lightweight TP-LMMSG without using HHBlits achieves the highest accuracy (0.7394±0.008), AUC (0.8025±0.0076), F1 score (0.6708±0.03) and SN (0.5824±0.0468). The heavyweight version that incorporates HMM features achieves the highest MCC (0.5089±0.0177). Taking the lightweight model as a reference, our model exhibits a 2.2% higher accuracy and AUC compared with the SOTA model, while the F1 and MCC scores demonstrate an improvement of over 4%. Moreover, the outstanding sensitivity of our model demonstrates a better ability to identify positive samples in a balanced dataset. The comparison also shows that, although AMPpred-MFA and AMPpred-EL achieved improved accuracy and AUC score compared with amPEPpy, they do not surpass sAMPpred-GAT model in terms of AMP prediction.

Besides the XUAMP dataset, our model on the remaining six datasets constantly exhibits promising performance (Figure 2). Given no significant difference observed between sAMPpred-GAT(full_nrdb) and sAMPpred-GAT(nrdb90), we adopt the latter for subsequent evaluations. The box plots reveal that our model outperforms the previous methods on every dataset. By strictly following the feature extraction procedure of the amPEPpy approach, we find that the amPEPpy model exhibits a similar AUC score on the APD3 dataset compared with AMPpred-MFA and sAMPpred-GAT. However, the amPEPpy model struggles with performance bias on several datasets, including dbAMP, DRAMP and LAMP. Nevertheless, our approaches demonstrate stable performance advantages across multiple datasets, highlighting the robustness of our method across diverse AMP data. The

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Table 2. The performance comparison of TP-LMMGS with the other 13 AMP prediction models on XUAMP test set

Method	ACC	AUC	F1	MCC	SN	SP	Source
AMPscannerV2	0.568	0.585	0.548	0.137	0.523	0.613	[3]
iAMP-2L	0.592	0.592	0.368	0.261	0.238	0.947	[3]
ADAM-SVM	0.612	0.612	0.471	0.264	0.346	0.878	[3]
ampir	0.563	0.619	0.379	0.156	0.266	0.859	[3]
MLAMP	0.553	0.629	0.230	0.194	0.133	0.972	[3]
ADAM-HMM	0.684	0.684	0.623	0.390	0.521	0.847	[3]
AMPlify	0.642	0.697	0.462	0.381	0.307	0.976	[3]
AMPEP	0.658	0.727	0.487	0.425	0.325	**0.992**	[3]
AMPfun	0.674	0.735	0.555	0.414	0.406	0.943	[3]
amPEPpy	0.679	0.742	0.554	0.433	0.398	0.960	This work
AMPpred-MFA	0.687	0.750	0.630	0.395	0.532	0.842	This work
AMPpred-EL	0.697	0.766	-	0.437	0.480	0.913	[45]
sAMPpred-GAT(full_nrdb)	0.712	0.779	0.643	0.459	0.529	0.894	This work
sAMPpred-GAT(nrdb90)	0.716	0.780	0.649	0.466	0.528	0.903	This work
TP-LMMSG(HMM)	0.7342±0.0088	0.8001±0.0073	0.6708±0.03	**0.5089±0.0177**	0.547±0.0627	0.9208±0.0531	This work
TP-LMMSG(no_HMM)	**0.7394±0.0077**	**0.8025±0.0076**	**0.6895±0.0226**	0.5059±0.0134	**0.5824±0.0468**	0.8959±0.0377	This work

Note: All results generated in this study are obtained from 10 repetitions with random model initialization. The **bold** and underlined data refer to the best and second best value of each evaluation metric.

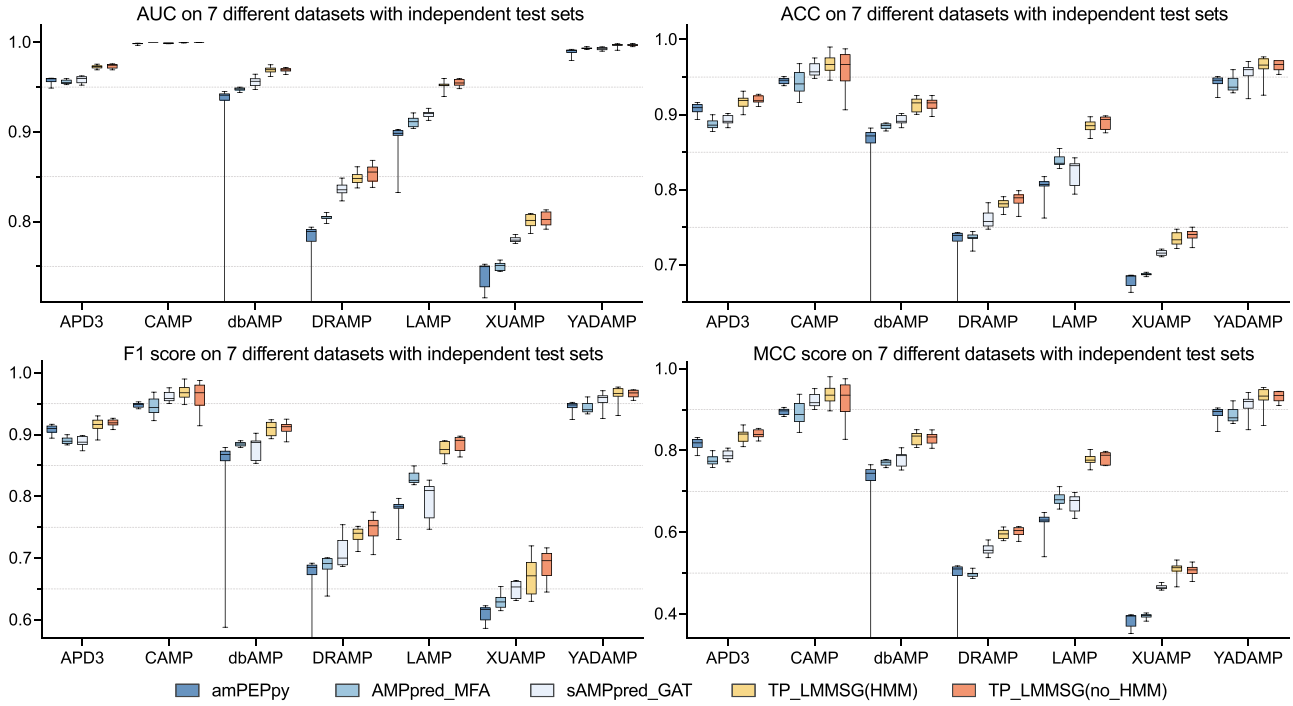

Figure 2. Performance comparison among the amPEPpy, AMPpred-MFA, sAMPpred-GAT and two versions of TP-LMMSG models on seven AMP independent test sets. All results generated in this study are obtained from 10 repetitions with random model initialization.

detailed evaluation metrics tables of model performance on the six datasets are listed in Tables S2–S5.

Model interpretation with t-SNE

In this section, we analyze the differences in feature extraction among the three deep learning approaches: AMPpred-MFA, sAMPpred-GAT (nrdb90) and TP-LMMSG (no_HMM). The learned features from the fully connected layer preceding the classification layer of each model are extracted and projected into 2D views (Figure 3 and Figure S3).

Through a horizontal comparison of the above three methods on seven datasets, the two methods with graph neural networks share similar feature distributions. On the other hand, the method AMPpred-MFA with LSTM layer separate the samples in a different manner. Compared with the other methods, TP-LMMSG illustrates a more uniform feature separation through all seven datasets. The positive samples' feature projection demonstrates a higher level of concentration in the feature space. Moreover, the TP-LMMSG model also exhibits a remarkable separation in the border region between positive and negative samples. As for the AMPpred-MFA model, the decision boundary of the positive and negative samples are not clear. The sAMPpred-GAT model demonstrates superior performance compared with the AMPpred-MFA model, while still encounters similar challenges. However, for TP-LMMSG model, the incorporation of abundant node-level information help alleviate this issue. The t-SNE visualization of TP-LMMSG could provide insights into the enhanced positive sample prediction during previous evaluations.

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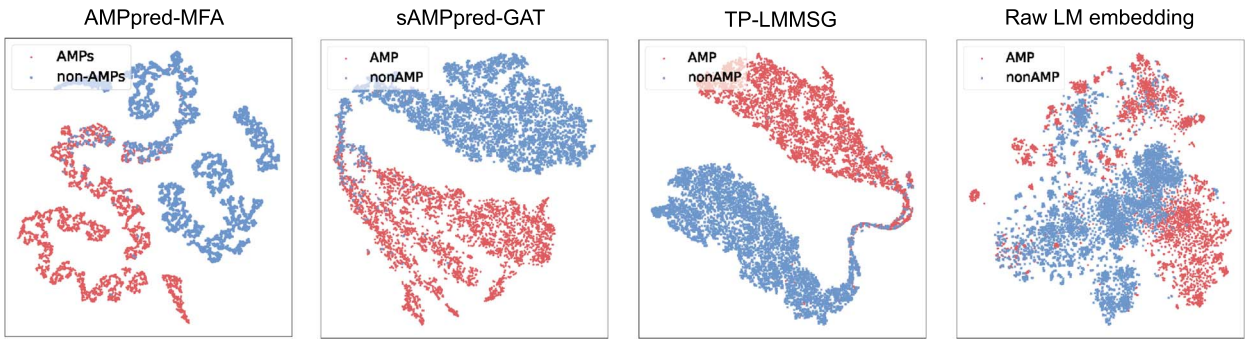

Figure 3. The t-SNE visualization of deep learning models on XUAMP dataset. From left to right is AMPpred-MFA, sAMPpred-GAT, TP-LMMSG and raw language model output embedding.

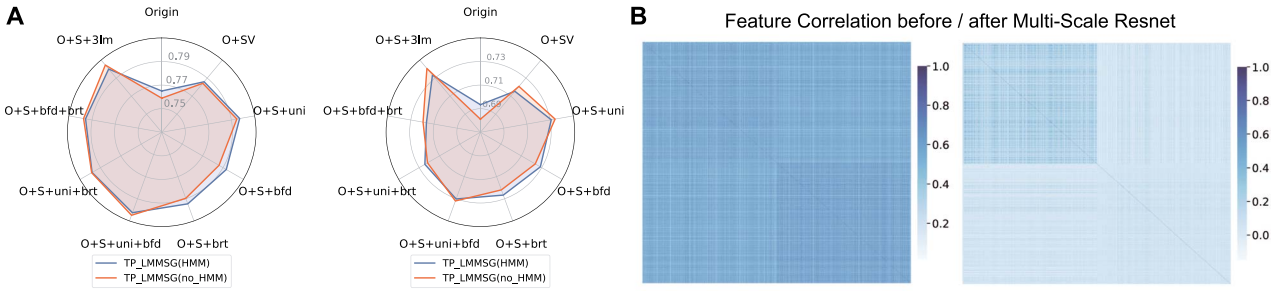

Figure 4. The result of ablation experiment, 10-fold cross-validation and time comparison. (**A**) The ablation experiment results for TP-LMMSG (HMM) and TP-LMMSG (no_HMM) with different LM feature. (**B**) The LM embedding correlation before and after the flexible HMSRN module.

In addition to the three deep learning model, the raw language model embeddings is also projected onto the 2D views. It appears that the embeddings of the language model exhibit a certain degree of discrimination between positive and negative samples. Specifically, LM embeddings demonstrate a more concentrated distribution for negative samples, while the embeddings of positive samples appear to be scattered in the feature space.

Ablation analysis

To further investigate how LM embeddings intrinsically influence our model, we aim to assess the impact of different features for peptide prediction. To achieve this, we conduct ablation experiments to examine the effect of each language model embedding. Specifically, for the four different pre-trained language models involved in this study, we first evaluate the effect of SeqVec embeddings. The amino acid level feature generated by 'uni', 'bfd' and 'brt' is subsequently incorporated into the model, resulting in a total of nine different feature combinations (Figure 4A). For each feature setting, we have also examined the influence of removing HMM features. A 10-fold cross-validation is conducted on the XUAMP dataset for objective evaluation with each feature combination (Figure S4).

According to the radar chart of AUC and accuracy score, incorporating the language model generated feature indeed improves the overall performance of our model. With an increasing number of incorporated language models, there is a progressive enhancement in the model's performance. Due to the absence of HMM features, when incorporating only up to two language model features, TP-LMMSG (no_HMM) exhibits lower accuracy and AUC score compared with TP-LMMSG (HMM). However, as more features are introduced, the impact of HMM features becomes less pronounced. The best performance of TP-LMMSG (no_HMM) is achieved when incorporating all four LM embeddings. Furthermore, we observe that the use of different language model features also has an impact on the model's performance. The combination of 'uni-bfd' and 'uni-brt' features yield better results compare to the 'bfd-brt' combination. Specifically, the decreasing performance of 'uni-brt' and 'bfd-brt' compared with 'uni-bfd' has demonstrated the importance of the ProtT5-XL-UniRef50 generated feature to our model.

Besides the ablation cross-validation, we extracted the original language model embeddings and the HMSRN output features to investigate the effectiveness of HMSRN in extracting peptide features. We randomly sample 10% of peptide sequences among all positive and negative samples. The feature correlation between each sequence is then computed and visualized with heat maps (Figure 4B). According to the heat map, the first half of the X and Y axes represent the positive samples, while the second half represents the negative samples. It can be observed that the initial LM embeddings do not exhibit significant feature correlation. Only a slight correlation can be observed among the negative samples. However, the HMSRN output illustrates an obvious correlation pattern before merging with the peptide graph nodes. These results further support the improved positive sample detection of TP-LMMSG model.

Time consumption comparisons

In this section, we assess the advantages of our model in preprocessing compared with the previous sAMPpred-GAT method. In light of the variations in total sample sizes and sequence lengths across different datasets, we compare the pre-processing time of TP-LMMSG and sAMPpred-GAT on four different datasets, which are XUAMP, APD3, CAMP and dbAMP. The average duration is then calculated for comparison. The time comparison is carried on a Intel Core 13700K CPU (Figure S5). The average pre-processing time for sAMPpred-GAT based on dataset XUAMP, APD3, CAMP and dbAMP are 15.1, 15.0, 15.2 and 15.1 s, respectively. In contrast,

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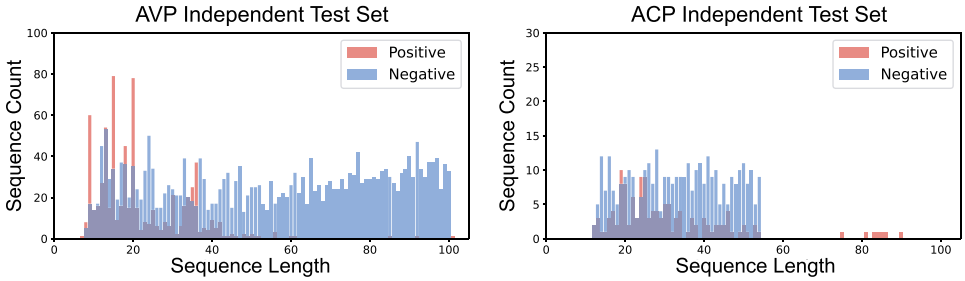

Figure 5. The distribution of peptide sequence and sequence length of the independent test datasets for the AVP dataset and ACP datset.

Table 3. The performance comparison among TP-LMMSG with the other three AVP prediction model

AVP	ACC	AUC	AUPR	MCC	SN	SP	Source
ENNAVIA	0.939	0.930	-	0.870	0.947	0.927	[4]
AVPIden	0.902	0.968	0.979	0.804	0.881	0.930	[4]
FFMAVP	0.924	0.962	0.977	0.846	0.917	0.933	[4]
TP-LMMSG	**0.9715±0.0141**	**0.9965±0.002**	**0.9883±0.0062**	**0.9211±0.0365**	**0.9812±0.0116**	**0.9688±0.0172**	This work

Note: Results generated in this study are obtained with 10-fold cross-validation. The **bold** data refer to the best value of each evaluation metric.

Table 4. The performance comparison among TP-LMMSG with the other three ACP prediction model

ACP	ACC	AUC	MCC	SN	SP	Source
mACPpred	0.870	0.899	0.659	0.879	0.838	[5]
iDACP	0.895	0.953	0.777	0.882	0.919	[5]
ACP-ML	0.926	0.975	0.843	**0.937**	0.908	[5]
TP-LMMSG	**0.9492±0.0162**	**0.9865±0.0093**	**0.9016±0.0292**	0.917±0.0436	**0.9814±0.0179**	This work

Note: Results generated in this study are obtained with 10-fold cross-validation. The **bold** data refer to the best value of each evaluation metric.

by eliminating HMM encoding, our method can compress the average feature acquisition duration into around 2 s. Specifically, for each sample in the four datasets, the time consumption is 2.1, 2.0, 2.2 and 2.2 s, respectively. This result demonstrates that our model can learn sequence features within a reasonable duration, thus improving the applicability of graph neural networks in predicting peptide properties.

Generalization ability assessment on AVP and ACP prediction tasks

Through the previous experiments, we have validated the effectiveness of the TP-LMMSG model in AMP prediction. Moreover, the model has exhibited robustness when applied to different independent test datasets. Hence, we aim to generalize the model to other therapeutic peptide prediction tasks to further evaluate the generalization ability of our model. The two tasks involved in this section are AVPs prediction and ACPs prediction. We employ the AVP dataset and the ACP dataset provided in previous literature [4, 5]. Different from the AMP dataset, the AVPs and ACPs dataset are both imbalanced. The sample distribution is listed in Table 1 and Figure 5. We conduct a new round of training for the TP-LMMSG model based on those two datasets. The model's performance on each dataset is evaluated through 10-fold cross-validation.

According to the result of AVP prediction, the TP-LMMSG model demonstrates an advantage over existing classification models (Table 3). It is worth noting that the previously proposed models only exhibit advantages across certain evaluation metrics. For example, the ENNAVIA model shows a higher accuracy, MCC and Sn score, while the AVPIden model exhibits a better AUC and AUPR score. The recently published FFMAVP model has the best Sp, while its other metrics are lower than the previous models. Nevertheless, the TP-LMMSG model achieves superiority across all metrics in the AVP prediction task. Compared with the existing models, the TP-LMMSG model demonstrates over 3% higher accuracy, AUC, Sn and Sp. Moreover, the MCC score is observed for an increase of 5%. The high AUPR values further suggest that our model exhibits strong performance even in the presence of imbalanced data.

In the ACP prediction tasks, the TP-LMMSG model also demonstrated its superiority (Table 4). In the independent test sets, TP-LMMSG outperforms the existing SOTA models in terms of metrics of interest: accuracy (2.3%), AUC (1.1%), MCC (5.8%) and Sp (7.3%). In contrast to the previous AMP datasets, the AVP and ACP datasets have sequence lengths ranging from 7 to 121 and 5 to 255, respectively. It demonstrates that our prediction model exhibits robust generalization across datasets with diverse distributions of sequence lengths.

Given the fact that positive and negative samples of therapeutic peptides show highly imbalanced distribution in the real world, we conduct a series of experiments on the XUAMP dataset with shrinking ratios of positive samples in the training dataset, thus to discover the performance boundary of peptide prediction models (Figure S6). Despite a noticeable decrease in the models' predictive performance when the positive samples in training set are decreased to a proportion of 10%, TP-LMMSG still demonstrates reliable predictive performance compared with the other methods. Considering that the positive-to-negative sample ratios of AVP and ACP training datasets are approximately 0.27 and 0.33 respectively, TP-LMMSG continues to showcase its performance within this range.

Discussion and conclusion

In this study, we develop a new model that resolves the time-consuming pre-processing challenge encountered by previous

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graph neural network models. Meanwhile, our model outperforms the current SOTA AMP prediction models across all datasets. Additionally, the TP-LMMSG also demonstrates robustness on imbalanced datasets, surpassing the current SOTA AVP and ACP prediction models.

Previous studies have primarily relied on MSA and various amino acid properties as the feature representation [3, 36]. The residue features, k-mer features and overall characteristics of peptide sequences can be obtained from sequences, thus facilitating the peptide property prediction. On the other hand, graph neural networks can leverage structural information of peptide molecules, but face challenges in specific amino acid property feature representations. Moreover, due to the limited number of homologs for partial peptides, running MSA may not yield any informative sequence information. In practice, neither nrdb90 nor entire nrdb guarantees completed MSA results for all sequences. This may introduce biases in feature extraction. With the node-level language model embeddings, our model can incorporate a large number of amino acid property representations, making it possible to integrate amino acid property features into a graph neural network.

In addition, the optimization in pre-processing stage enables the practical application of graph neural networks for peptide property prediction tasks. Compared with the previous graph learning model, our approach achieves a pre-processing time reduction of over 7 folds. Therefore, even when applied to larger databases with fine-tuning, our approach yields highly accurate results within reasonable time-frames. Moreover, our method could save a large amount of local storage space, enabling efficient local fine-tuning and adaptation to hardware resources at the small-scale laboratory level.

The good performance and potential exhibited by the TP-LMMSG model have sparked deeper thinking for peptide property prediction. It is acknowledged that the distribution of therapeutic peptides with clinical potential is extremely imbalanced compared with negative samples in the real world. Our experimental results also explore the performance boundary. This series of experiments with imbalanced datasets demonstrates the robustness of TP-LMMSG. It also suggests that the continuous supplements of novel positive samples should be enabled and included in benchmark datasets to reflect the practical application context.

Taking it a step further, with the advancement of biosynthetic technology, the introduction of non-natural amino acids is emerging in peptide drug design [46]. In the future, peptide graphs that involve non-natural amino acids can be constructed through site-directed mutagenesis and the incorporation of distinct node and edge features. By utilizing Density Functional Theory derived features or traditional small molecule representations, the limitations of protein language models could be expanded in peptide prediction tasks [47]. These avenues hold promise for further investigation. Moreover, mounting evidence suggests that covalently attached small molecules to peptides demonstrate potential drug-like properties, posing challenges in molecule-peptide graph constructions [48, 49]. Fortunately, with the development of protein prediction models, AlphaFold3 has acquired the capability to predict drug–protein interaction poses, enhancing the expansion possibilities of our workflow [50]. While AlphaFold3 is not open-sourced during the writing of this article, exploring cutting-edge protein structure prediction models provides a valuable focus for peptide property annotations in the future.

Key Points

- A generalizable peptide prediction model, TP-LMMSG, is proposed. It outperforms current state-of-the-art models in antimicrobial peptides, antiviral peptides, and anticancer peptides predictions.
- TP-LMMSG effectively tackles the challenge faced by previous graph neural network-based methods, which is the inability to extract and represent amino acid level properties.
- The model utilizes optimized amino acid level feature extraction strategy for rapid deployment of graph neural networks in peptide prediction.
- TP-LMMGS can flexibly incorporate novel language model embeddings. The architecture could providing continuous contributions to the discovery process of potential peptide therapeutics.

Supplementary data

Supplementary data is available at *Briefings in Bioinformatics* online.

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Data and Code Availability

The source code of TP-LMMGS and all the dataset we use are available at https://github.com/NanjunChen37/TP_LMMSG.

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